AP PRECALCULUS

Unit 1 • Worksheet 1.3A

Worksheet 1.3A • Rates of Change in Linear and Quadratic Functions

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Each list gives outputs at $x = 0, 1, 2, 3$. Which list could come from a linear function?

(A) $3, 5, 9, 15$

(B) $4, 7, 10, 13$

(C) $1, 2, 4, 8$

(D) $2, 3, 5, 8$

Question 2

NO CALCULATOR

Each list gives outputs at $x = 0, 1, 2, 3, 4$. Which list could come from a quadratic function?

(A) $1, 3, 9, 27, 81$

(B) $5, 8, 11, 14, 17$

(C) $0, 2, 8, 18, 32$

(D) $1, 2, 4, 7, 13$

Question 3

NO CALCULATOR

For $f(x) = ax^2 + bx + c$, the second differences of $f(0), f(1), f(2), \ldots$ are all -4. What is a?

(A) -4

(B) 4

(C) -2

(D) 2

Question 4

NO CALCULATOR

The inputs in a table are $0, 2, 4, 6$ and the outputs of a quadratic function have second differences equal to 8. What is a?

(A) 1

(B) 2

(C) 4

(D) 8

Question 5 (Free response) NO CALCULATOR

x	-1	0	1	2	3	4
y	9	4	1	0	1	4

(a) Show that the data can be modeled by a quadratic function.
(b) Find the quadratic function.
(c) On which interval is the function decreasing? How do the first differences show this?